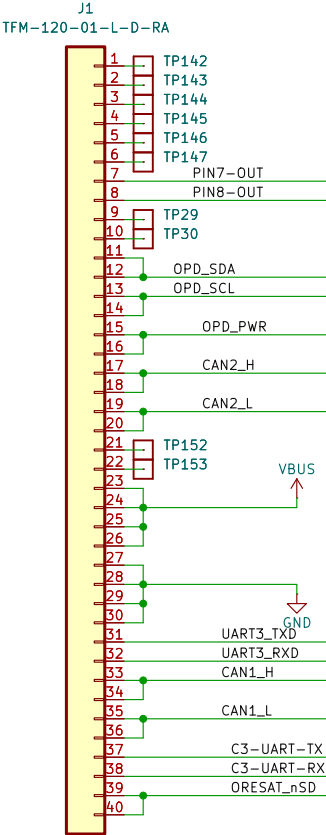
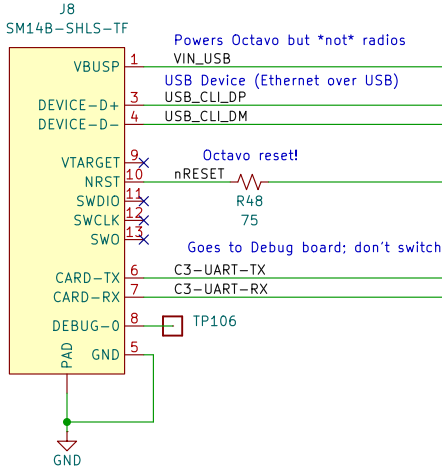


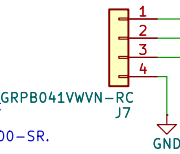
OreSat Card Main Connector  
(Backplane v2)



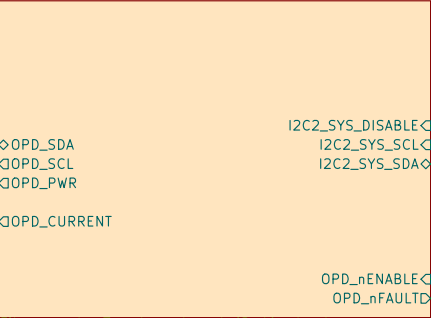
OreSat Card Debug Connector  
20 pin FFC Connector



Needed for programming  
boot EEPROM  
TPS65217 I2C Addr = 0x24  
EEPROM I2C Addr = 0x50  
Needs 4 pos 1.27 mm socket  
and cable, e.g.,  
Samtec SFSS-04-28-H-06.00-SR.



OPD and I2C Peripherals

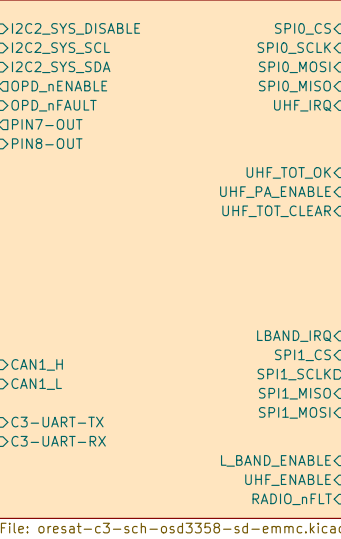


The Octavo OSD3358 schematics are heavily based on the Octavo Systems  
OSD3358-SM-RED platform, which seems to be released under an MIT-like license.  
Please see <https://octavosystems.com/docs/osd3358-sm-red-schematics-pdf/>

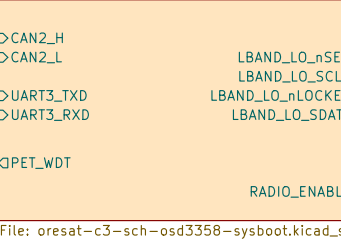
The OSD3358 design is also based on the beagleboard.org PocketBeagle.  
<https://github.com/beagleboard/pocketbeagle>

Bill of Materials is kept in a Google Sheet Here:  
<https://docs.google.com/spreadsheets/d/11vG1kWyRAjmbF5QJM-EhXmvQoF6047japwYlHF802do/edit#gid=668351130>

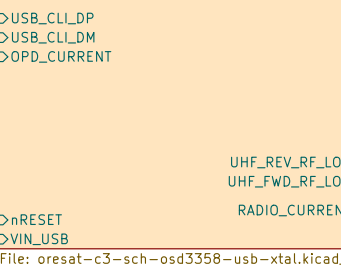
OSD3358: SIP C



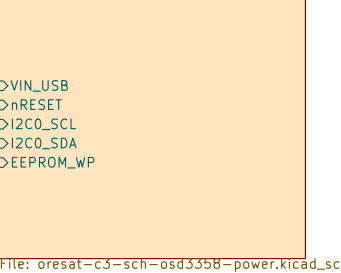
OSD3358: SIP D



OSD3358: SIP B



OSD3358: SIP A



OSD3358: SIP E + F

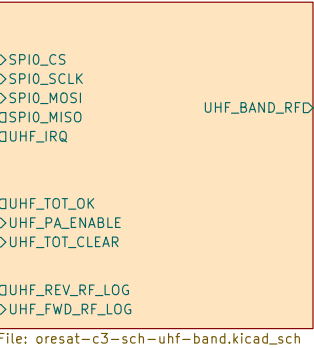


File: oresat-c3-sch-osd3358-sip-e-f.kicad\_sch

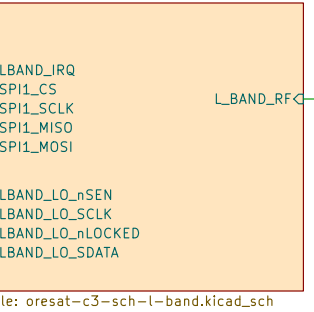
Oresat Card RF Connectors

SMMP Connectors to Backplane

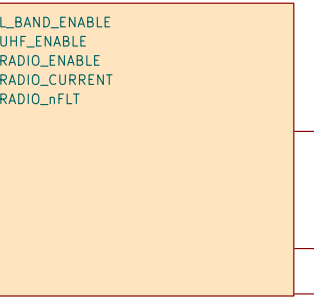
UHF Band



L Band



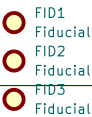
Power



File: oresat-c3-sch-06.kicad\_sch

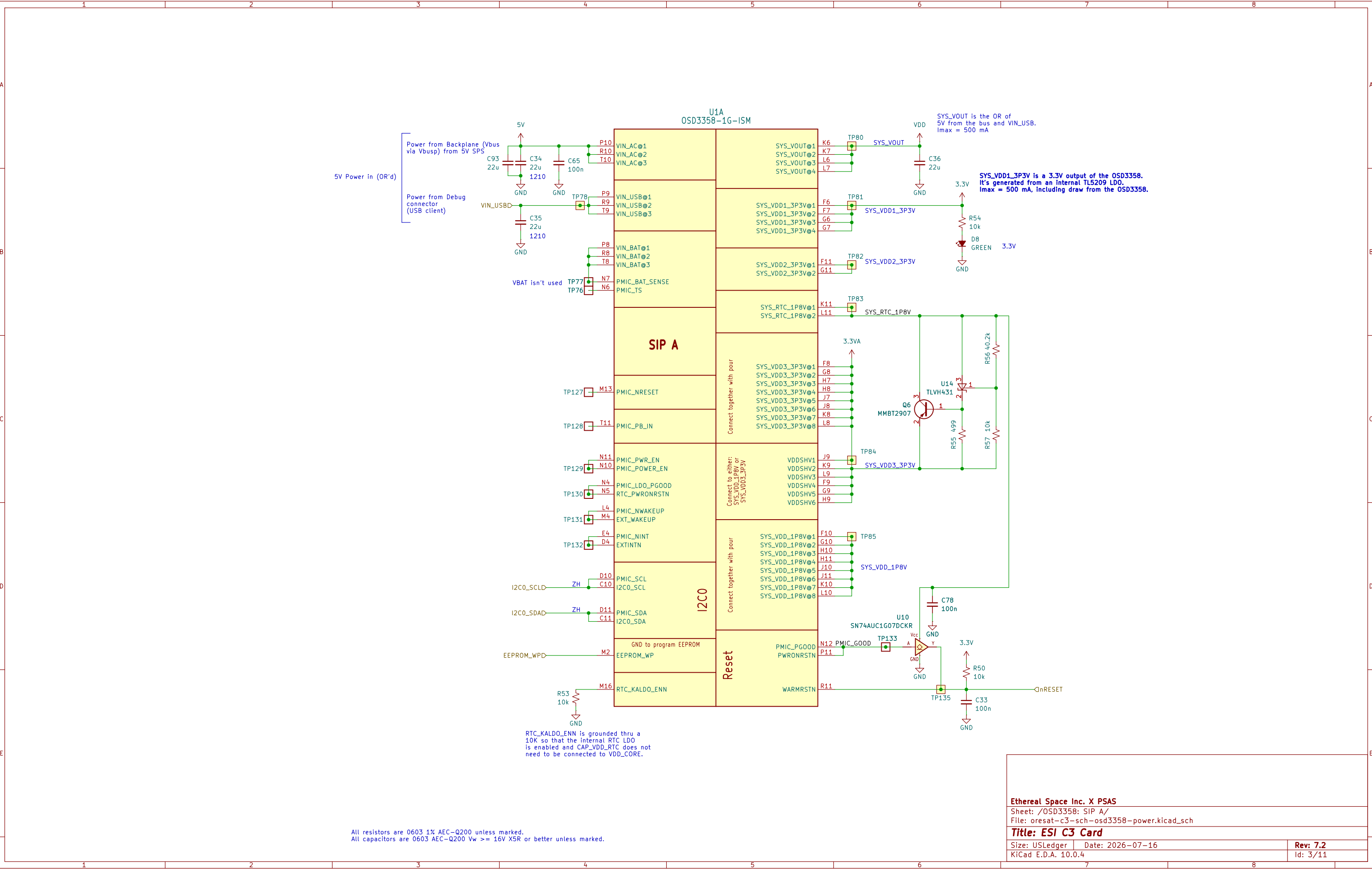
Mechanical

OreSat Card v1.3 with RF Connectors 2,3



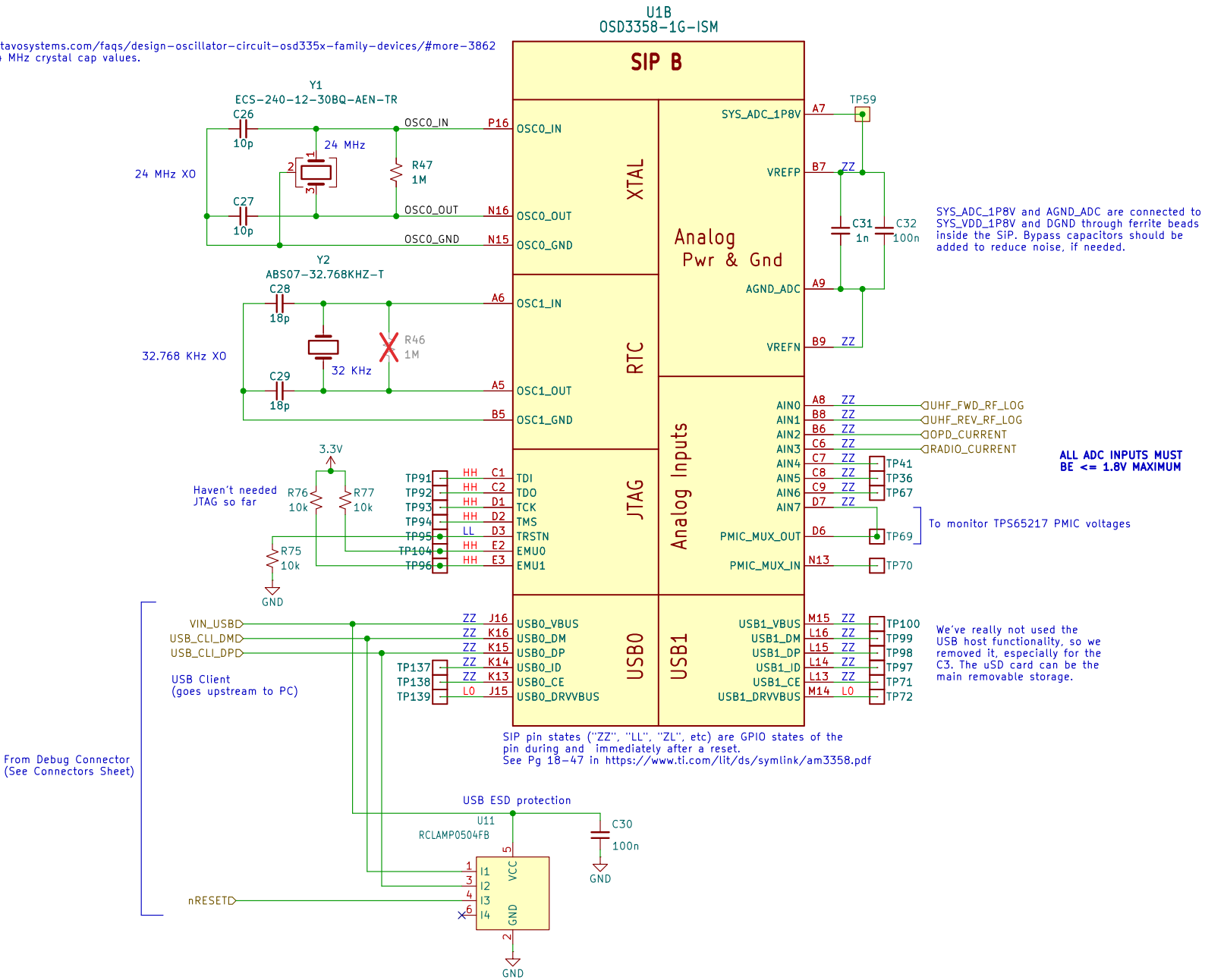
Fiducials  
(Front)





All resistors are 0603 1% AEC-Q200 unless marked.  
All capacitors are 0603 AEC-Q200 Vw >= 16V X5R or better unless marked.

See <https://octavosystems.com/faqs/design-oscillator-circuit-osd335x-family-devices/#more-3862> to calculate 24 MHz crystal cap values.



All resistors are 0603 1% AEC-Q200 unless marked.  
All capacitors are 0603 AEC-Q200 Vw >= 16V X5R or better unless marked.

Ethereal Space Inc. X PSAS  
Sheet: /OSD3358: SIP B/  
File: oresat-c3-sch-osd3358-usb-xtal.kicad\_sch

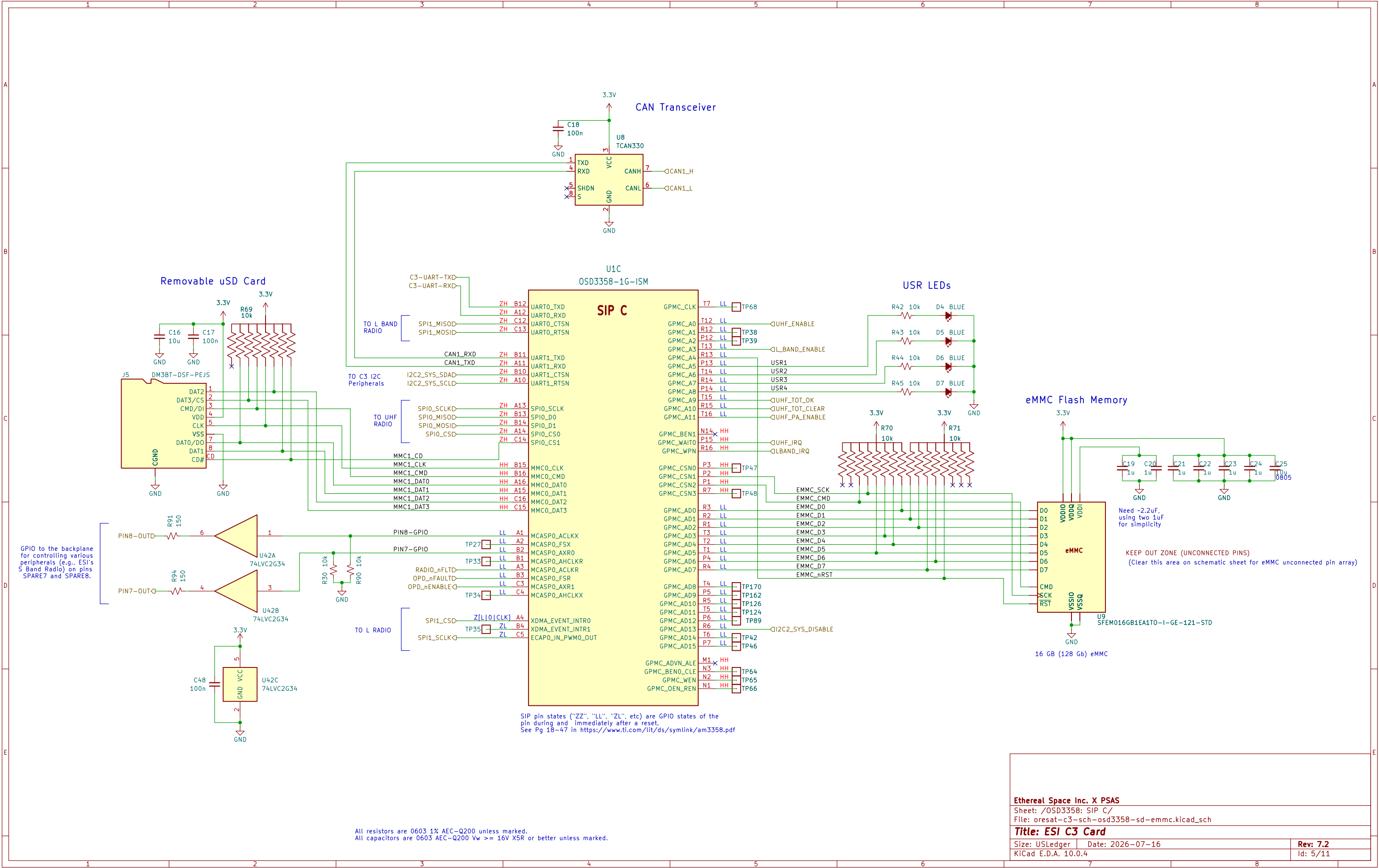
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Size: USLedger Date: 2026-07-16

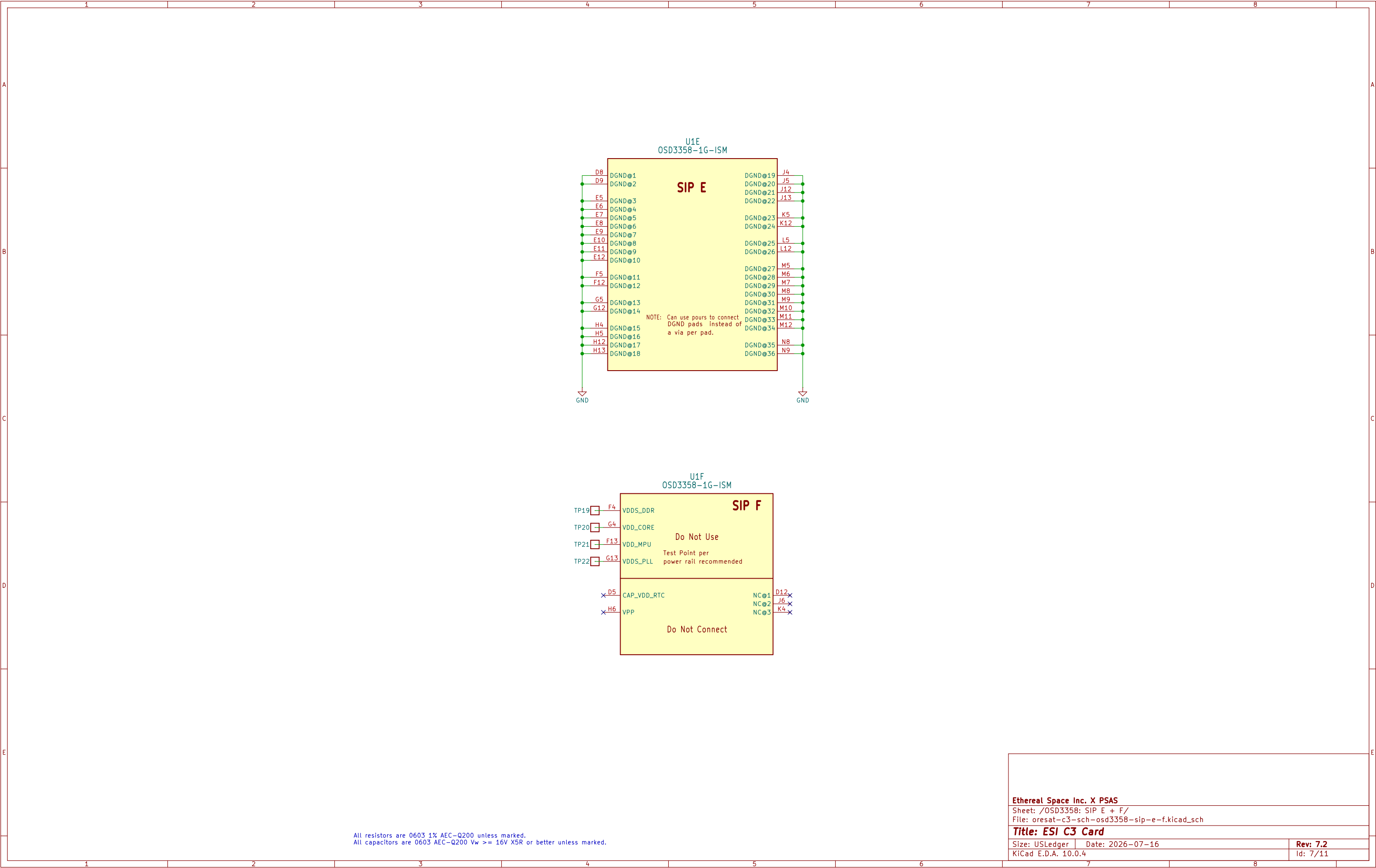
Rev: 7.2

KiCad E.D.A. 10.0.4

Id: 4/11







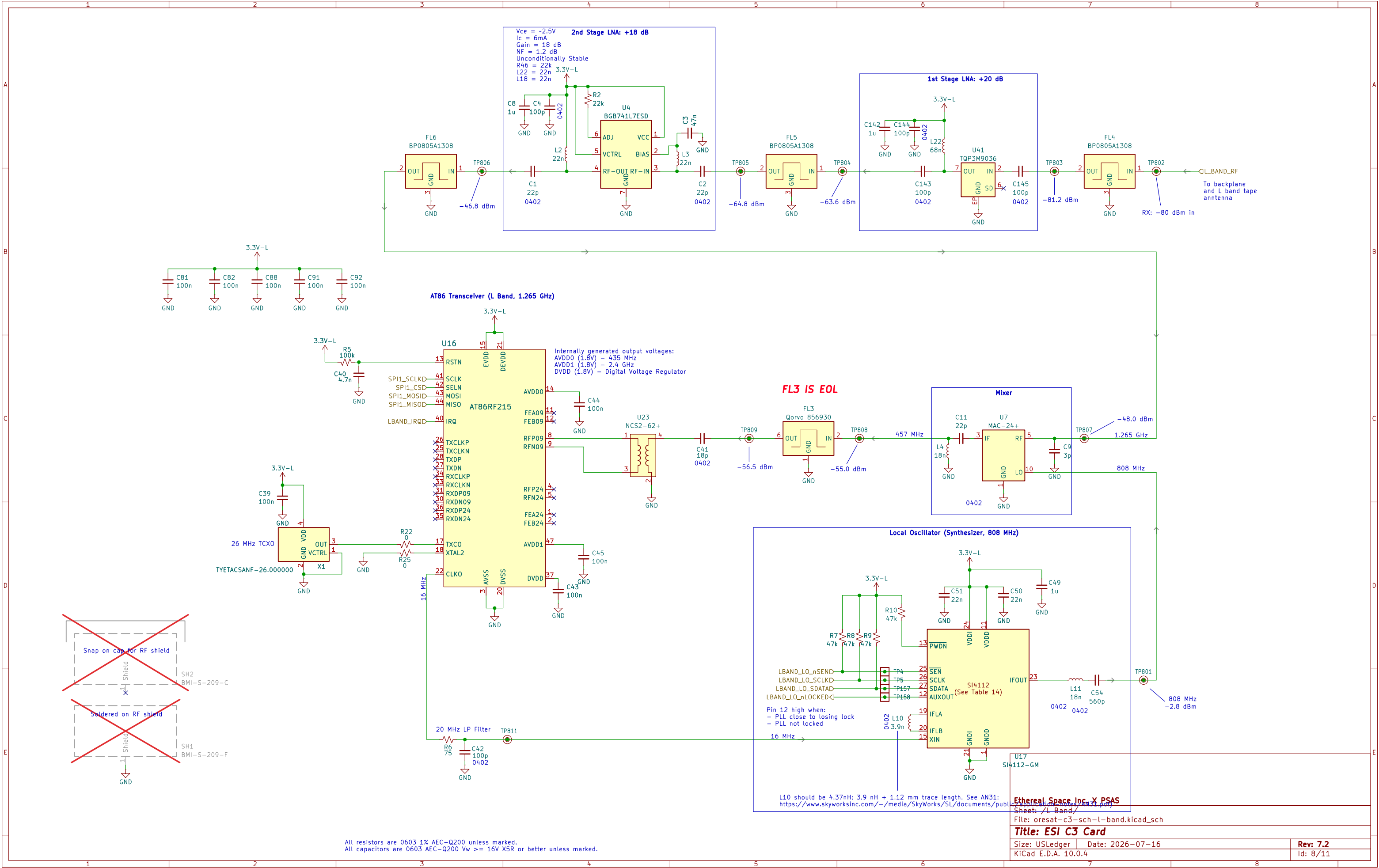
All resistors are 0603 1% AEC-Q200 unless marked.  
All capacitors are 0603 AEC-Q200 Vw >= 16V X5R or better unless marked.

Ethereal Space Inc. X PSAS

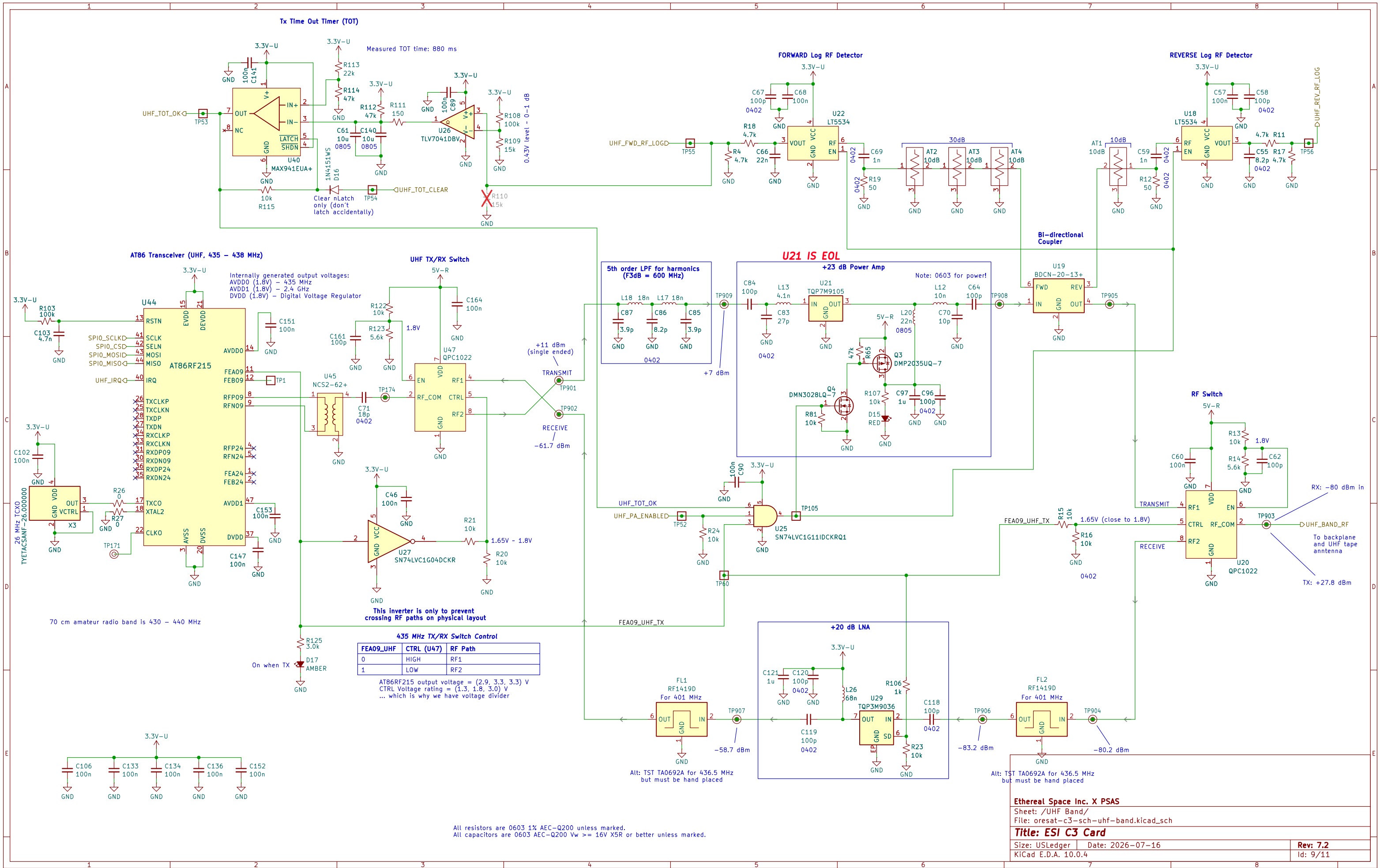
Sheet: /OSD3358: SIP E + F/  
File: oresat-c3-sch-osd3358-sip-e-f.kicad\_sch

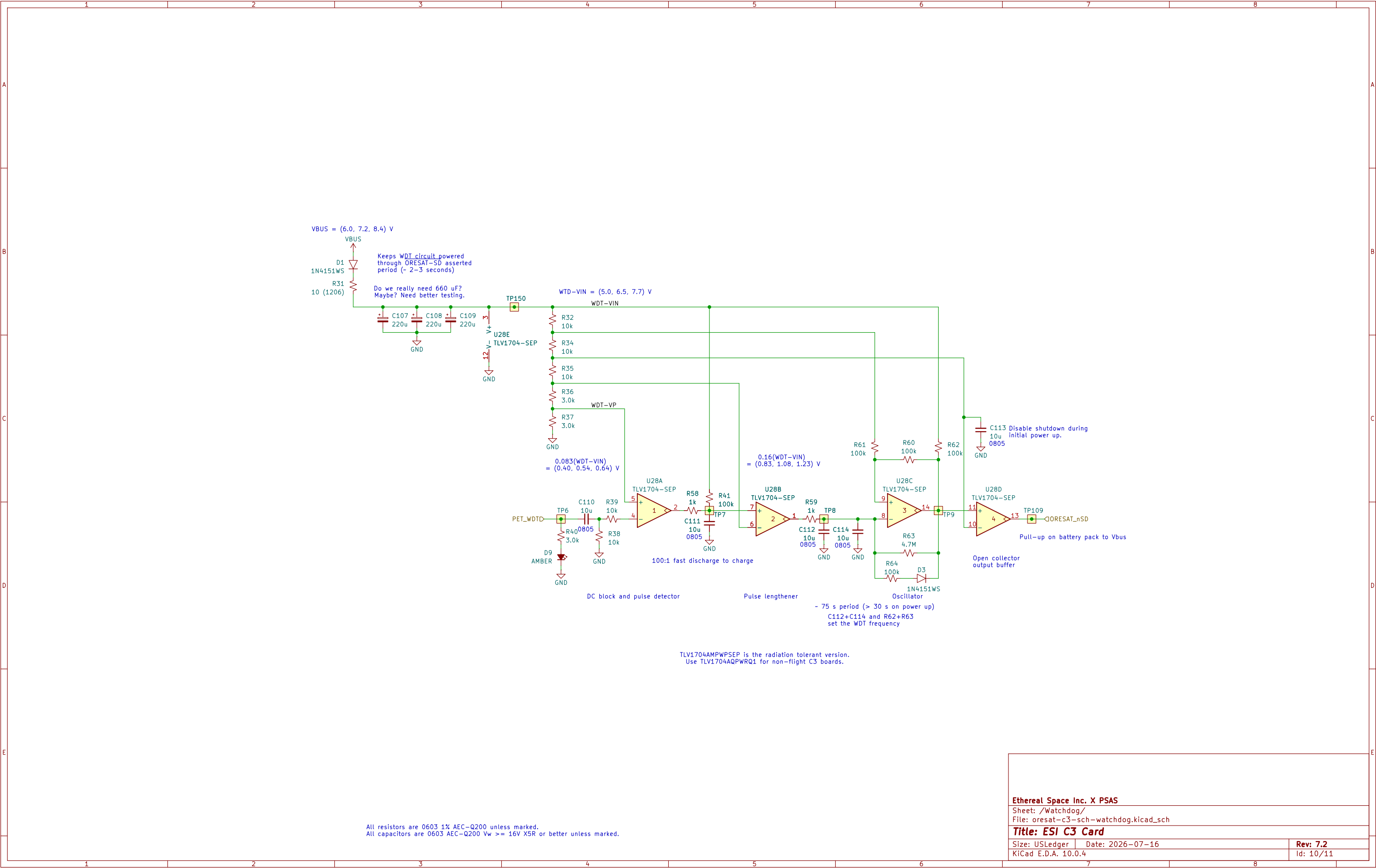
Title: ESI C3 Card

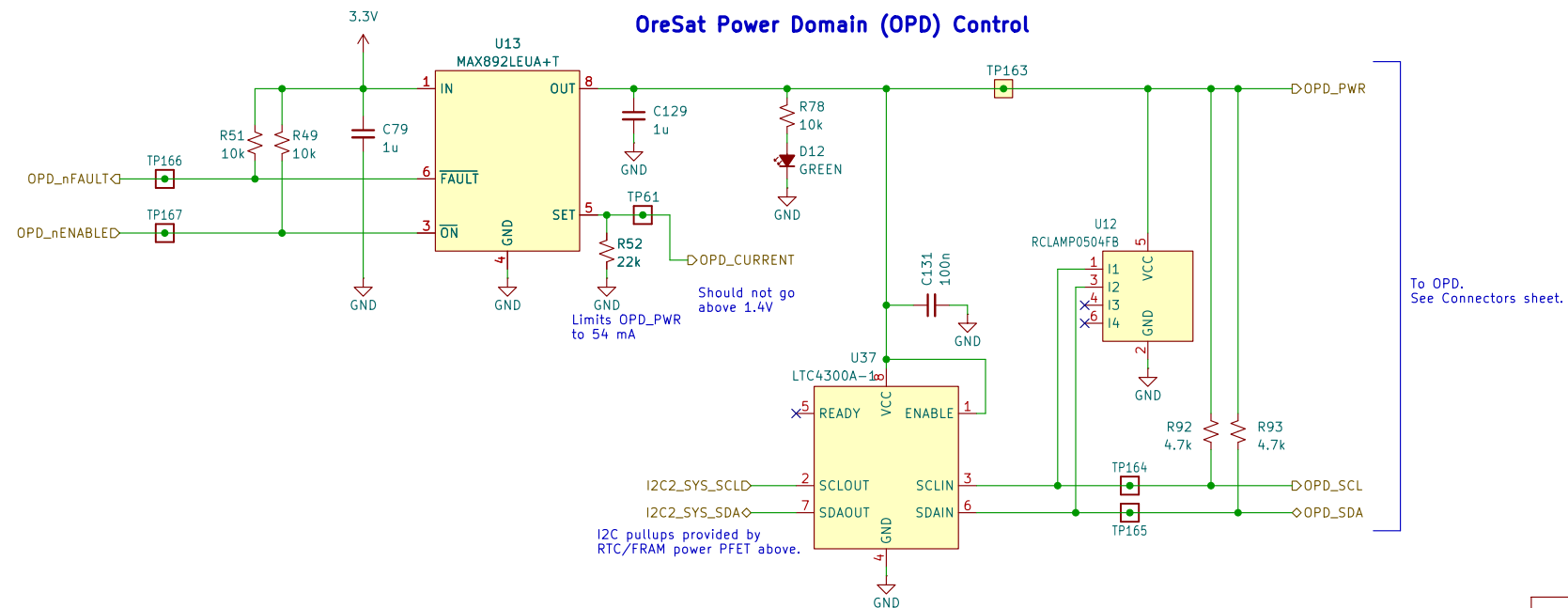
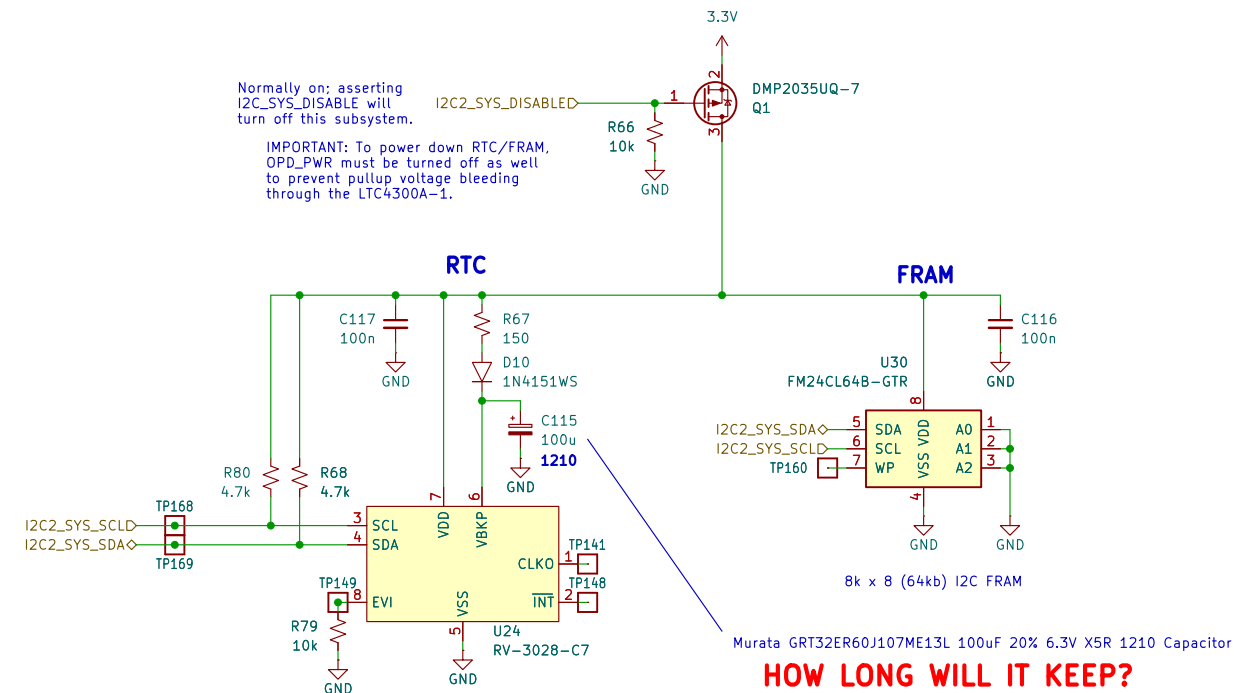
Size: USLedger    Date: 2026-07-16    Rev: 7.2  
KiCad E.D.A. 10.0.4    Id: 7/11



All resistors are 0603 1% AEC-Q200 unless marked.  
All capacitors are 0603 AEC-Q200 Vw >= 16V X5R or better unless marked.







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